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MATHS MASTER _____

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CANDIDATE NUMBER

2024 Trial HSC Examination

Form VI Mathematics Extension 2

Tuesday 13th August 2024

8:40am

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Total Marks: 100

Section I (10 marks) Questions 1 – 10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11 – 16

- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on each booklet and on the multiple choice sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 6 booklets per boy
- Candidature: 77 pupils

Writer: PC

Section I

Questions in this section are multiple-choice.

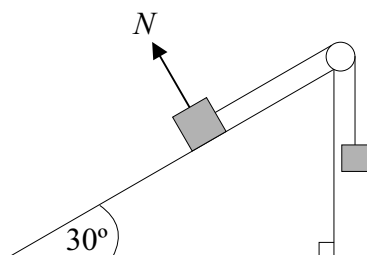
Record the single best answer for each question on the provided answer sheet.

1. In the Argand diagram, the complex number z lies in the second quadrant. In which quadrant does the complex number $i\bar{z}$ lie?
 - (A) first
 - (B) second
 - (C) third
 - (D) fourth
2. Given $\underline{u} = \underline{i} + 2\underline{j}$, $\underline{v} = \underline{j} + 3\underline{k}$, what is the projection of $\underline{u}$ onto $\underline{v}$?
 - (A) $\frac{1}{5}(\underline{i} + 2\underline{j})$
 - (B) $\frac{1}{5}(\underline{j} + 3\underline{k})$
 - (C) $\frac{7}{10}(\underline{i} + 2\underline{j})$
 - (D) $\frac{7}{10}(\underline{j} + 3\underline{k})$
3. A particle is moving in Simple Harmonic Motion with amplitude 3 metres. Its speed is 4 metres per second when the particle is 1 metre from the centre of motion. What is the period of the motion?
 - (A) $\frac{\pi}{2}$
 - (B) $\frac{\pi}{\sqrt{2}}$
 - (C) $\sqrt{2}\pi$
 - (D) 2π
4. Consider the identity: $\frac{8}{(x+1)(x-1)^2} \equiv \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.
Which of the following are the correct values of A , B and C ?
 - (A) $A = 2$, $B = -2$, $C = 4$
 - (B) $A = 2$, $B = -2$, $C = -4$
 - (C) $A = -2$, $B = 2$, $C = 4$
 - (D) $A = -2$, $B = 2$, $C = -4$

5. Let $\overrightarrow{OP} = \frac{1}{2}(\sqrt{2}\underline{i} - \underline{j} + \underline{k})$, and α , β and γ be the angles that $\overrightarrow{OP}$ makes with the positive x , y and z -axes respectively. What is the value of $\alpha + \beta + \gamma$?

- (A) 45°
 (B) 165°
 (C) 180°
 (D) 225°

6.



Two masses are attached to a light inextensible string which is looped over a smooth pulley as shown in the diagram. The larger mass is on a smooth incline of 30° to the horizontal, and the smaller mass (M kg) hangs freely. The masses are stationary and at equilibrium, and the magnitude of the acceleration due to gravity is $g \text{ m/s}^2$. What is the magnitude, in Newtons, of the normal reaction force N , on the larger mass?

- (A) $N = \frac{\sqrt{3}}{2}Mg$
 (B) $N = Mg$
 (C) $N = \sqrt{3}Mg$
 (D) $N = 2Mg$

7. Given that $|z| = 2$, what is the greatest possible value of $\text{Arg}(z + 4i)$?

- (A) $\frac{\pi}{6}$
 (B) $\frac{\pi}{3}$
 (C) $\frac{2\pi}{3}$
 (D) $\frac{5\pi}{6}$

8. A single die is rolled and the uppermost face noted. Let p represent the statement “the uppermost face is divisible by 3” and let q represent the statement “the uppermost face is divisible by 6”. Considering the implication $p \Rightarrow q$, which of the following is correct?
- (A) The negation is true and the converse is true.
 - (B) The negation is true and the converse is false.
 - (C) The negation is false and the converse is true.
 - (D) The negation is false and the converse is false.
9. If w is the complex root of $z^5 = 1$ with smallest positive argument, which of the following is false?
- (A) $\operatorname{Re}(w + w^3) < 0$
 - (B) $\operatorname{Im}(w + w^3) > 0$
 - (C) $\operatorname{Re}(w + w^4) > 0$
 - (D) $\operatorname{Im}(w + w^4) < 0$
10. Given that x and y are real numbers, which of the following is a true statement?
- (A) $\forall y, \exists x$ such that $x^2 - y^2 = x$
 - (B) $\forall y, \exists x$ such that $x^2 - y^2 = y$
 - (C) $\forall y, \exists x$ such that $x^2 + y^2 = x$
 - (D) $\forall y, \exists x$ such that $x^2 + y^2 = y$

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet.

Marks

(a) Consider two complex numbers $z = a + 2i$ and $w = 1 - ai$, where a is real.

(i) Find zw in the form $x + iy$.

1

(ii) Find $z - aw$ in modulus-argument form.

1

(iii) Show that $(\overline{w})^2 + 2w$ is real.

1

(b) Find the indefinite integrals:

(i) $\int \frac{e^x}{1 + e^{2x}} dx$

1

(ii) $\int \sin^4 x \sin 2x dx$

1

(c) Sketch the region in the complex plane where the inequalities $\operatorname{Re}(z) < 1$, $\operatorname{Re}(z) < \operatorname{Im}(z)$, and $-\frac{\pi}{2} < \operatorname{Arg}(z + 1) < \frac{\pi}{4}$ all hold simultaneously.

3

(d) A particle moving along the x -axis has acceleration a , velocity v and displacement x at time t . Initially, $x = 0$ and $v = 2$.

(i) If $v = x^2 + 1$, find a when $x = 3$.

2

(ii) If $a = x^2 + 1$, find v when $x = 3$.

3

(e) Consider the complex numbers u , v and z , such that $u = 2i$, $|v| = 3$ and $z = uv$. Find the exact value of $|z - v|$.

2

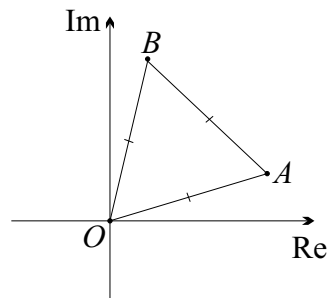
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QUESTION TWELVE (15 marks)

Start a new answer booklet.

Marks

(a)



Let O , A and B be points in the complex plane representing the numbers 0 , $6 + 2i$ and z . If $\triangle OAB$ is equilateral, with vertices in anti-clockwise order, determine the exact value of z in the form $x + iy$. 2

(b) (i) Use the substitution $t = \tan \frac{x}{2}$ to show that $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x} = \ln 2$. 2

(ii) Hence use the substitution $x = \frac{\pi}{2} - u$, to find the exact value of 3

$$\int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \cos x + \sin x}.$$

(c) Given $z^4 - 2z^3 + 9z^2 - 6z + 18 = 0$ has $1 + i\sqrt{5}$ as a root, find all the roots. 3

(d) Consider the line l with equation $\underline{r} = \begin{bmatrix} 7 \\ 4 \\ 13 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 7 \\ 10 \end{bmatrix}$ and the sphere S with equation 3

$$\left| \underline{r} - \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \right| = 3. \text{ Show that } l \text{ touches } S \text{ and find the point of contact } Q.$$

(e) Prove, using contraposition, that $\forall x, y \in \mathbf{Z}$, if $x^2(y + 3)$ is even, then x is even or y is odd. 2

The paper continues on the next page

QUESTION THIRTEEN (15 marks)

Start a new answer booklet.

Marks

- (a) Use integration by parts twice to find $\int x^3(\log x)^2 dx$. 4
- (b) Show that $(\cos \theta + i \sin \theta)^n (\sin \theta + i \cos \theta)^n = e^{\frac{ni\pi}{2}}$. 2
- (c) (i) Show that $\frac{2}{(x+1)(x^2+1)} \equiv \frac{1}{x+1} - \frac{x-1}{x^2+1}$. 1
- (ii) Let $I_n = \int_0^1 \frac{2x^n}{(x+1)(x^2+1)} dx$.
- (α) Show that $I_0 = \frac{1}{2} \log 2 + \frac{\pi}{4}$. 2
- (β) By considering $I_0 + I_2$, or otherwise, find the exact value of I_2 . 2
- (d) (i) Give an example of positive integers m , n and p , where p is prime, such that $(2m+3)^2 = n^2 + p$. 1
- (ii) Use proof by contradiction to show that if p is prime and n is a positive integer, then no positive integer m exists such that $(5m+3)^2 = n^2 + p$. 3

The paper continues on the next page

QUESTION FOURTEEN (15 marks)

Start a new answer booklet.

Marks

- (a) Consider a typical point $R(1 - 2\lambda, 2 + 2\lambda, 3 - \lambda)$ on the line $l: \underline{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 2 \\ -1 \end{bmatrix}$,
and a typical point $Z(0, 0, \mu)$ on the z -axis, where λ and μ are non-zero parameters.

(i) Show that $\overrightarrow{RZ}$ is perpendicular to the z -axis when $\mu + \lambda = 3$. 1

(ii) Find the values of μ and λ such that $\overrightarrow{RZ}$ is perpendicular to both l and the z -axis. 2

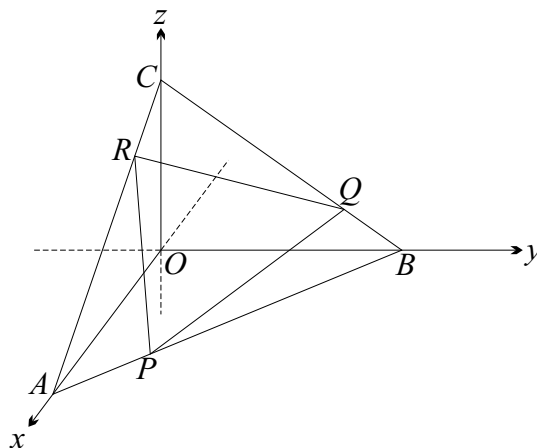
(iii) Hence find the shortest distance between l and the z -axis. 2

- (b) (i) Show that $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. 1

(ii) Hence, or otherwise, show that for positive integers a and b , 3

$$(a^7 + b^7)(a^2 + b^2) \geq (a^5 + b^5)(a^4 + b^4).$$

(c)



In the diagram, A , B and C lie on the positive x , y and z -axes respectively, and let $\overrightarrow{OA} = 4\underline{a}$, $\overrightarrow{OB} = 4\underline{b}$, $\overrightarrow{OC} = 4\underline{c}$, $\overrightarrow{AP} = \frac{1}{4}\overrightarrow{AB}$, $\overrightarrow{BQ} = \frac{1}{4}\overrightarrow{BC}$ and $\overrightarrow{CR} = \frac{1}{4}\overrightarrow{CA}$.

(i) Show that $\overrightarrow{PQ} = -3\underline{a} + 2\underline{b} + \underline{c}$. 2

(ii) It can also be shown that $\overrightarrow{QR} = -3\underline{b} + 2\underline{c} + \underline{a}$ (**do not** prove this). 2

Show that $\overrightarrow{PQ} \cdot \overrightarrow{QR} = -3|\underline{a}|^2 - 6|\underline{b}|^2 + 2|\underline{c}|^2$.

(iii) Given $|\underline{a}| = 2$ and $|\underline{b}| = 1$, show that if $\triangle PQR$ is right angled at Q , then it is also isosceles. 2

The paper continues on the next page

QUESTION FIFTEEN (15 marks) Start a new answer booklet.**Marks**

- (a) A projectile of unit mass is launched vertically upwards from the origin with an initial velocity of $\sqrt{3}g$ m/s, where g m/s² is the acceleration due to gravity. The projectile experiences a resistive force of magnitude $\frac{v^2}{g}$ Newtons, where v m/s is the velocity of the particle after t seconds. The acceleration of the projectile is given by

$$\ddot{x} = -g - \frac{v^2}{g}.$$

- (i) Show that $v = g \tan\left(\frac{\pi}{3} - t\right)$. 2
- (ii) Find an expression for the displacement x metres in terms of g and t . 2
- (iii) Show that the maximum height achieved by the particle is $g \ln 2$ metres. 1
- (iv) Derive an expression for x in terms of v^2 and show that this equation confirms the maximum height found in part (iii). 2

- (b) Consider the sequence of numbers: 1, -1, -5, -7, 1, 23, ...

These numbers can be generated using the recurrence relation

$$T_{n+2} = 2T_{n+1} - 3T_n, \text{ for } n \geq 1 \text{ with } T_1 = 1 \text{ and } T_2 = -1.$$

- (i) Use the recurrence relation to find T_7 . 1
- (ii) Show that the formula $T_n = \frac{(1 + i\sqrt{2})^n + (1 - i\sqrt{2})^n}{2}$ generates T_1 and T_2 . 2
- (iii) Use Mathematical Induction to prove the formula for T_n in part (ii) works for all positive integers n . 3
- (iv) Show that the formula for T_n in part (ii) is equivalent to 2

$$T_n = \left(\sqrt{3}\right)^n \cos\left(n \tan^{-1} \sqrt{2}\right).$$

The paper continues on the next page

QUESTION SIXTEEN (15 marks) Start a new answer booklet.**Marks**

- (a) (i) Using a suitable trigonometric substitution, or otherwise, show that

3

$$\int_0^1 \sqrt{x - x^2} dx = \frac{\pi}{8}.$$

- (ii) Given the integral
- $I_n = \int_0^1 x^{n+\frac{1}{2}} \sqrt{1-x} dx$
- , for integers
- $n \geq 0$
- , show that

2

$$I_n = \frac{2n+1}{2n+4} I_{n-1}.$$

- (iii) Show that for integers
- $n \geq 0$
- :

3

$$\int_0^1 x^n \sqrt{x - x^2} dx = \frac{(2n+1)! \pi}{2^{2n+2} (n+2)! n!}.$$

- (b) (i) Using de Moivre's Theorem, or otherwise, show that
- $\frac{\sin 2k\theta}{\sin \theta \cos \theta}$
- ,

3where k is a positive integer, can always be expressed as a polynomial in $\sin^2 \theta$.

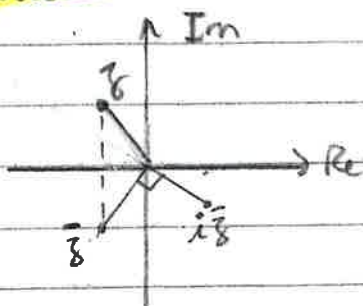
- (ii) Obtain the polynomial in
- $\sin^2 \theta$
- corresponding to
- $\frac{\sin 8\theta}{\sin \theta \cos \theta}$
- , and hence solve the equation:
- $z^6 - 6z^4 + 10z^2 - 4 = 0$
- .

4————— **END OF PAPER** —————

Section I

M/C

①



D

② $\text{Proj}_{\underline{u}} \underline{v} = \frac{1 \times 0 + 2 \times 1 + 0 \times 3}{1^2 + 2^2} (\underline{j} + 3\underline{k})$

$= \frac{1}{5} (\underline{j} + 3\underline{k})$

B

③ $v^2 = n^2 (a^2 - (x-c)^2)$

$4^2 = n^2 (3^2 - (1)^2)$

$n^2 = 2$

$n = \sqrt{2} \quad T = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$

C

④ $8 \equiv A(x-1)^2 + B(x+1)(x-1) + C(x+1)$

$x=1 \Rightarrow 8=2C \Rightarrow C=4$

$x=-1 \Rightarrow 8=4A \Rightarrow A=2$

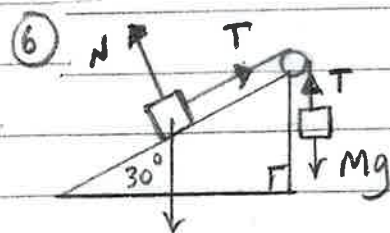
coeff of $x^2 \Rightarrow A+B=0 \Rightarrow B=-2$

A

⑤ $\alpha = \cos^{-1}(\frac{\sqrt{2}}{2}) = 45^\circ, \beta = \cos^{-1}(-\frac{1}{2}) = 120^\circ$

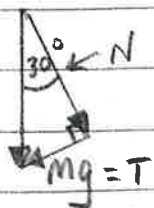
$\gamma = \cos^{-1}(\frac{1}{2}) = 60^\circ, \alpha + \beta + \gamma = 225^\circ$

D



Equilibrium $\Rightarrow$

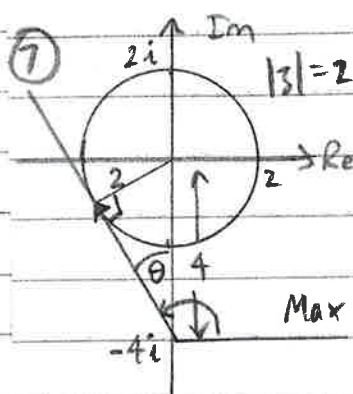
$T = Mg$



$\frac{N}{Mg} = \cot 30^\circ$

$N = \sqrt{3}Mg$

C



$\sin \theta = \frac{1}{2} = \frac{1}{2}$

$\theta = \frac{\pi}{6}$

Max Arg $(z+4i) = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$

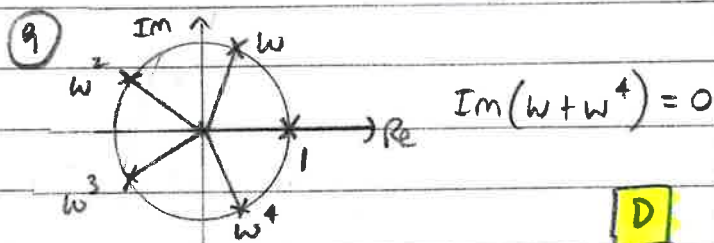
C

⑧ $p \Rightarrow q$ is false (3 uppermost)

Negation: $p \wedge \sim q$ True $\sim (p \Rightarrow q)$

Converse: $q \Rightarrow p$ True as 6 is div by 3.

A



D

⑩ $\forall y, \exists x : x^2 - y^2 = x$

$x^2 - x = y^2$

$(x - \frac{1}{2})^2 = y^2 + \frac{1}{4}$

ens $\geq \frac{1}{4} \forall y$, so always solutions for x .

Not the case for the others

A

Overall DBCADCCADA

Section II

11

(a) $z = a + 2i$, $w = 1 - ai$

(i) $zw = (a + 2i)(1 - ai)$
 $= a - a^2i + 2i + 2a$
 $= 3a + (2 - a^2)i$ ✓

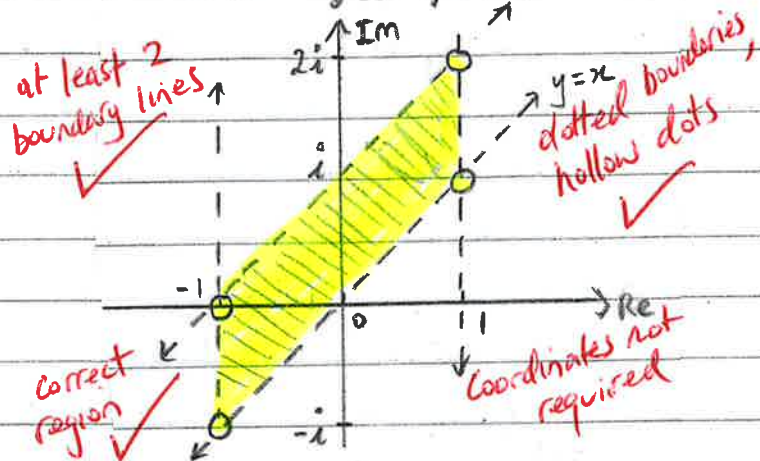
(ii) $z - aw = a + 2i - a(1 - ai)$
 $= a + 2i - a + a^2i$
 $= (a^2 + 2)i$
 $= (a^2 + 2) \operatorname{cis} \frac{\pi}{2}$ ✓

(iii) $(\bar{w})^2 + 2w = (1 - ai)^2 + 2(1 - ai)$
 $= 1 - 2ai - a^2 + 2 - 2ai$
 $= 3 - a^2$ ✓ (Real)

(b) (i) $\int \frac{e^x}{1 + e^{2x}} dx = \int \frac{e^x}{1 + (e^x)^2} dx$
 $= \tan^{-1}(e^x) + C$ ✓

(ii) $\int \sin^4 x \cos^2 x dx = \int \sin^4 x (2 \sin x \cos x) dx$
 $= 2 \int \sin^5 x \cos x dx$
 $= \frac{2}{6} \sin^6 x + C$ ✓

(c) $\operatorname{Re}(z) < 1$, $\operatorname{Im}(z) < \operatorname{Im}(z)$
 $-\frac{\pi}{2} < \operatorname{Arg}(z + 1) < \frac{\pi}{4}$



(d) $t = 0$, $x = 0$, $v = 2$

(i) $v = x^2 + 1$, $a = v \cdot \frac{dv}{dx}$
 $= (x^2 + 1)(2x)$ ✓

$x = 3 \Rightarrow a = (3^2 + 1)(2 \times 3)$
 $= 60$ ✓

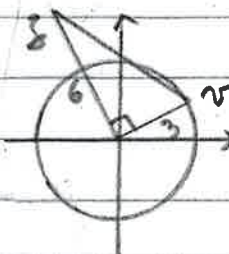
(ii) $a = \frac{d}{dx}(\frac{1}{2}v^2) = x^2 + 1$
 $\frac{1}{2}v^2 = \frac{1}{3}x^3 + x + C$ ✓

$t = 0$, $x = 0$, $v = 2 \Rightarrow C = 2$

$\downarrow$
 $a > 1$ so $v > 2$ $\Rightarrow v^2 = \frac{2}{3}x^3 + 2x + 4$ ✓ (positive only)
 $x = 3 \Rightarrow v = \sqrt{\frac{2}{3}(3)^3 + 2(3) + 4} = \sqrt{28} = 2\sqrt{7}$ ✓

(e)

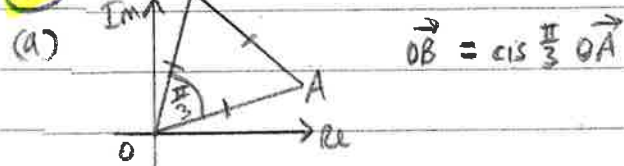
$|z - v| = |v||u - 1|$
 $= 3|-1 + 2i|$
 $= 3\sqrt{5}$



$u = 2i$, $|v| = 3$, $z = uv$

$|z - v| = \sqrt{3^2 + 6^2}$
 $= \sqrt{45} \text{ or } 3\sqrt{5}$ ✓

12



$$\vec{z} = \vec{OB} = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)(6+2i) \checkmark$$

$$= 3+i + 3\sqrt{3}i - \sqrt{3}$$

$$= (3-\sqrt{3}) + (1+3\sqrt{3})i \checkmark$$

(b) (i) $t = \tan \frac{x}{2} \Rightarrow x = \tan^{-1}(2t), dx = \frac{2dt}{1+t^2}$

$x=0, t=0$
 $x=\frac{\pi}{2}, t=1$

$$I = \int_0^1 \frac{2dt}{1+t^2 + 1-t^2 + 2t} \checkmark$$

$$= \int_0^1 \frac{dt}{1+t} = [\ln|1+t|]_0^1 = \ln 2 \checkmark$$

(ii) $x = \frac{\pi}{2} - u \Rightarrow dx = -du, x=0, u=\frac{\pi}{2}$
 $x=\frac{\pi}{2}, u=0$

$$I = \int_{\frac{\pi}{2}}^0 \frac{\left(\frac{\pi}{2} - u\right)(-du)}{1 + \cos\left(\frac{\pi}{2} - u\right) + \sin\left(\frac{\pi}{2} - u\right)} \checkmark$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} - u}{1 + \sin u + \cos u} du$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{du}{1 + \cos u + \sin u} - I \checkmark$$

$$\therefore 2I = \frac{\pi}{2} \ln 2 \quad (\text{from (i)})$$

$$I = \frac{\pi}{4} \ln 2 \checkmark$$

(c) $z^4 - 2z^3 + 9z^2 - 6z + 18 = 0$, Root: $1+i\sqrt{5}$
Since the coefficients are real, complex roots come in conjugate pairs, hence $1-i\sqrt{5}$ is also a root. Let the others be α, β .

Using sum and product of roots

$$\alpha + \beta + 1+i\sqrt{5} + 1-i\sqrt{5} = 2, \quad \alpha\beta(1+i\sqrt{5})(1-i\sqrt{5}) = 18$$

$$\alpha + \beta = 2, \quad \alpha\beta(1+5) = 18$$

$$\alpha = -\beta \quad (1) \quad \alpha\beta = 3 \quad (2)$$

both $\checkmark$

From (1) and (2) $\alpha^2 = -3 \Rightarrow \alpha = \pm\sqrt{3}i, \beta = \mp\sqrt{3}i$
Roots, $1 \pm \sqrt{5}i, \pm\sqrt{3}i \checkmark$

(d) $L: \begin{pmatrix} 7 \\ 4 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 7 \\ 10 \end{pmatrix}, S: \left| \begin{pmatrix} 7 \\ 4 \\ 13 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right| = 3$

Sub L into S

$$(7+2\lambda-3)^2 + (4+7\lambda+1)^2 + (13+10\lambda-2)^2 = 9$$

$$(4+7\lambda)^2 + (5+7\lambda)^2 + (11+10\lambda)^2 = 9 \checkmark$$

$$16 + 16\lambda + 4\lambda^2 + 25 + 70\lambda + 49\lambda^2 + 121 + 220\lambda + 100\lambda^2 = 9$$

$$153\lambda^2 + 306\lambda + 153 = 0$$

$$\lambda^2 + 2\lambda + 1 = 0$$

$$(\lambda+1)^2 = 0, \lambda = -1 \checkmark$$

one solution, vector of Q: $(7+2(-1), 4+7(-1), 13+10(-1))$
 $\therefore Q$ is $(5, -3, 3) \checkmark$

(e) RTP $\forall x, y \in \mathbb{Z}$, if $x^2(y+3)$ is even, then x is even or y is odd.

Contrapositive: RTP If x is odd and y is even then $x^2(y+3)$ is not even (ie odd) $\checkmark$

Let $x = 2k+1, y = 2l, k, l \in \mathbb{Z}$

then $x^2(y+3) = (2k+1)^2(2l+3)$
 $= (4k^2+4k+1)(2l+3)$
 $= 8k^2l + 12k^2 + 4kl + 12k + 2l + 3$
 $= 2(4k^2l + 6k^2 + 2kl + 6k + l + 1) + 1$
which is odd.

Hence proven.

(13) (a) $I = \int x^3 (\log x)^2 dx$ Let $u = (\log x)^2$, $v' = x^3$
 $u' = 2 \log x \left(\frac{1}{x}\right)$, $v = \frac{1}{4} x^4$

$I = \frac{1}{4} x^4 (\log x)^2 - \frac{1}{2} \int x^3 \log x dx$
 Let $U = \log x$, $V' = x^3$
 $U' = \frac{1}{x}$, $V = \frac{1}{4} x^4$

$I = \frac{1}{4} x^4 (\log x)^2 - \frac{1}{2} \left(\frac{1}{4} x^4 \log x - \frac{1}{4} \int x^3 dx \right)$
 $= \frac{1}{4} x^4 (\log x)^2 - \frac{1}{8} x^4 \log x + \frac{1}{32} x^4 + C$
 $= \frac{1}{32} x^4 [8(\log x)^2 - 4 \log x + 1] + C$

(b) RTP $(\cos \theta + i \sin \theta)^n (\sin \theta + i \cos \theta)^n = e^{\frac{n i \pi}{2}}$

LHS = $[(\cos \theta + i \sin \theta)(\sin \theta + i \cos \theta)]^n$
 $= [\cos \theta \sin \theta + i \cos^2 \theta + i \sin^2 \theta - \sin \theta \cos \theta]^n$
 $= (i)^n$
 $= (e^{i \frac{\pi}{2}})^n$
 $= e^{\frac{n i \pi}{2}} = \text{RHS}$

(c) (i) RTP $\frac{2}{(x+1)(x^2+1)} = \frac{1}{x+1} - \frac{x-1}{x^2+1}$

RHS = $\frac{x^2+1 - (x-1)(x+1)}{(x+1)(x^2+1)}$
 $= \frac{x^2+1 - x^2+1}{(x+1)(x^2+1)}$
 $= \frac{2}{(x+1)(x^2+1)} = \text{LHS}$

(ii) $I_n = \int_0^1 \frac{2x^n}{(x+1)(x^2+1)} dx$

(d) $I_0 = \int_0^1 \frac{2}{(x+1)(x^2+1)} dx$

$I_0 = \int_0^1 \left(\frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} \right) dx$ from (i)

$= \left[\ln|x+1| - \frac{1}{2} \ln|x^2+1| + \tan^{-1} x \right]_0^1$
 $= \ln 2 - \frac{1}{2} \ln 2 + \frac{\pi}{4} - (0 - \frac{1}{2}(0) + 0)$
 $= \frac{1}{2} \ln 2 + \frac{\pi}{4}$

(b) $I_0 + I_2 = \int_0^1 \frac{2(1+x^2)}{(x+1)(x^2+1)} dx$

$= [2 \ln|x+1|]_0^1$
 $= 2 \ln 2 - 0$
 $= 2 \ln 2$

$\therefore I_2 = 2 \ln 2 - \left(\frac{1}{2} \ln 2 + \frac{\pi}{4} \right)$
 $= \frac{3}{2} \ln 2 - \frac{\pi}{4}$

(d) (i) $(2m+3)^2 = n^2 + p$ true for
 $m=2, n=6, p=13$ or equivalent

(ii) Assume $\exists m \in \mathbb{Z}^+ : (5m+3)^2 = n^2 + p$,
 where $n \in \mathbb{Z}^+$, p is prime.

So $p = (5m+3)^2 - n^2$
 $= (5m+3+n)(5m+3-n)$

If p is prime, $(5m+3-n) = 1$ (the smaller factor)
 So $5m+2 = n$

And $p = (5m+3+n) = (5m+3+5m+2)$
 $= 5(2m+1)$

which is not prime.

Hence a contradiction.

$\therefore \exists \text{ no } m \in \mathbb{Z}^+ : (5m+3)^2 = n^2 + p$

14) $\lambda: \vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}, R(1-2\lambda, 2+2\lambda, 3-\lambda)$
 $Z(0, 0, \mu)$

(i) $\vec{RZ} \cdot \vec{r} = 0 \Rightarrow \begin{pmatrix} -1+2\lambda \\ -2-2\lambda \\ \mu-3+\lambda \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ \mu \end{pmatrix} = 0$

$\mu \neq 0$
 $\lambda \neq 0$
 $\mu(\mu-3+\lambda) = 0 \Rightarrow \mu + \lambda = 3$ (1)

(ii) $\vec{RZ} \cdot \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} = 0 \Rightarrow -2+4\lambda-4+4\lambda-\mu+3-\lambda=0$

$\mu - 9\lambda = -1$ (2)

(1) + (2) $\Rightarrow -8\lambda = 2 \Rightarrow \lambda = -\frac{1}{4}, \mu = \frac{13}{4}$

(iii) R is $(1-2(-\frac{1}{4}), 2+2(-\frac{1}{4}), 3-(-\frac{1}{4}))$
 $u(\frac{3}{2}, \frac{3}{2}, \frac{13}{4}), Z(0, 0, \frac{13}{4})$

Shortest Distance = $\sqrt{(\frac{3}{2})^2 + (\frac{3}{2})^2 + 0^2}$
 $= \frac{3}{\sqrt{2}} \text{ or } \frac{3\sqrt{2}}{2} \text{ units}$

(b) (i) RTP $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

LHS = $a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3$

$= a^3 - b^3$

$= \text{RHS}$

(ii) RTP $(a^7 + b^7)(a^4 + b^4) \geq (a^5 + b^5)(a^4 + b^4)$

LHS - RHS = $a^9 + a^7b^2 + a^5b^4 + b^9 - (a^9 + a^5b^4 + a^4b^5 + b^9)$

$= a^7b^2 - a^4b^5 - a^5b^4 + a^2b^7$

$= a^4b^2(a^3 - b^3) - a^2b^4(a^3 - b^3)$

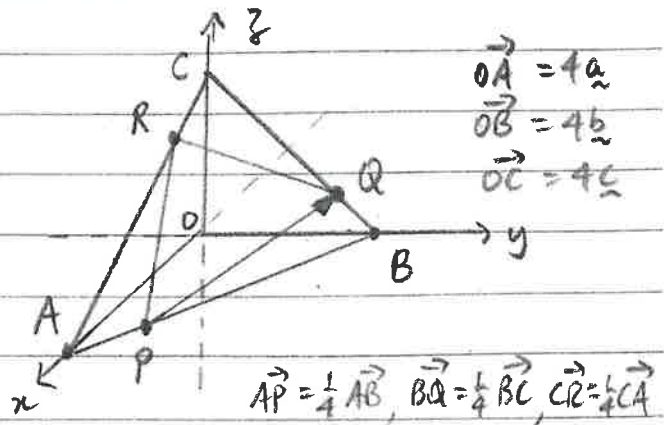
$= (a^3 - b^3)a^2b^2(a^2 - b^2)$

$= (a-b)(a^2 + ab + b^2)a^2b^2(a+b)(a-b)$

$= a^2b^2(a-b)^2(a+b)(a^2 + ab + b^2)$

≥ 0 for $a, b \in \mathbb{Z}^+$ so proven

(c)



(i) $\vec{AP} = \frac{1}{4}(4b - 4a) = b - a$

$\vec{OP} = \vec{OA} + \vec{AP} = 4a + b - a = 3a + b$

$\vec{BQ} = \frac{1}{4}(4c - 4b) = c - b$

$\vec{OQ} = \vec{OB} + \vec{BQ} = 4b + c - b = 3b + c$

$\therefore \vec{PQ} = \vec{OQ} - \vec{OP} = 3b + c - (3a + b) = -3a + 2b + c$

(ii) Given $\vec{QR} = -3b + 2c + a$

$\vec{PQ} \cdot \vec{QR} = (-3a + 2b + c) \cdot (-3b + 2c + a)$

$= 9a \cdot b - 6a \cdot c - 3|a|^2 - 6|b|^2 + 4b \cdot c$

$+ 2a \cdot b - 3b \cdot c + 2|c|^2 + a \cdot c$

$= -3|a|^2 - 6|b|^2 + 2|c|^2$ since

$a \cdot b = b \cdot c = a \cdot c = 0$

(axes perpendicular)

(iii) If $\angle PQR = 90^\circ$, $\vec{PQ} \cdot \vec{QR} = 0$, $|a| = 2, |b| = 1$

$\Rightarrow 0 = -3(2)^2 - 6(1)^2 + 2|c|^2$

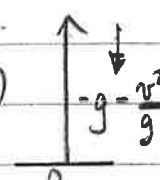
$\Rightarrow 18 = 2|c|^2 \Rightarrow |c|^2 = 9 \Rightarrow |c| = 3$

So $a = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, b = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, c = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$

$\vec{PQ} = \begin{pmatrix} -6 \\ 2 \\ 3 \end{pmatrix}, |\vec{PQ}|^2 = 36 + 4 + 9 = 49$

$\vec{QR} = \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix}, |\vec{QR}|^2 = 4 + 9 + 36 = 49$

$\therefore |\vec{PQ}| = |\vec{QR}| = 7$, $\triangle PQR$ is isosceles.

15 (a)  $t=0, x=0, v=\sqrt{3}g$
 $\ddot{x} = -g - \frac{v^2}{g}$

(i) $\frac{dv}{dt} = -\frac{g^2+v^2}{g}$

$t = -g \int \frac{1}{g^2+v^2} dv$

$= -\tan^{-1}\left(\frac{v}{g}\right) + C_1$ ✓

$t=0, v=\sqrt{3}g \Rightarrow C_1 = \tan^{-1}\sqrt{3} = \frac{\pi}{3}$

$t = \frac{\pi}{3} - \tan^{-1}\left(\frac{v}{g}\right)$

$\tan^{-1}\left(\frac{v}{g}\right) = \frac{\pi}{3} - t$

$v = g \tan\left(\frac{\pi}{3} - t\right)$ ✓ ①

(ii) $x = \int g \tan\left(\frac{\pi}{3} - t\right) dt$

$= g \int \frac{\sin\left(\frac{\pi}{3} - t\right)}{\cos\left(\frac{\pi}{3} - t\right)} dt$

$= g \ln|\cos\left(\frac{\pi}{3} - t\right)| + C_2$ ✓

$t=0, x=0 \Rightarrow C_2 = -g \ln|\cos\frac{\pi}{3}|$

$= -g \ln\left(\frac{1}{2}\right)$

$= g \ln 2$

$\therefore x = g \ln\left[2 \cos\left(\frac{\pi}{3} - t\right)\right]$ ② ✓

(iii) $v=0$ in ① $\Rightarrow t = \frac{\pi}{3}$ then in ②

$x = g \ln 2 \cos 0$

$= g \ln 2$ Max height. ✓

(iv) $v \cdot \frac{dv}{dx} = -\frac{g^2+v^2}{g}$

$x = \int \frac{-g v}{g^2+v^2} dv$

$= -\frac{g}{2} \ln|g^2+v^2| + C_3$ ✓

$x=0, v=\sqrt{3}g \Rightarrow C_3 = \frac{g}{2} \ln 4g^2$

So $x = \frac{g}{2} \ln\left(\frac{4g^2}{g^2+v^2}\right)$

and $v=0 \Rightarrow x = \frac{g}{2} \ln 4 = \frac{g}{2} \ln 2^2 = g \ln 2$ ✓

(b) 1, -1, -5, -7, 1, 23, ...

$T_{n+2} = 2T_{n+1} - 3T_n$, $T_1=1, T_2=-1$

(i) $T_7 = 2T_6 - 3T_5$

$= 2 \times 23 - 3 \times 1$

$= 43$ ✓

(ii) $T_n = \frac{1}{2} \left((1+i\sqrt{2})^n + (1-i\sqrt{2})^n \right)$

$T_1 = \frac{1}{2} \left((1+i\sqrt{2})^1 + (1-i\sqrt{2})^1 \right) = 1$ ✓

$T_2 = \frac{1}{2} \left((1+i\sqrt{2})^2 + (1-i\sqrt{2})^2 \right)$

$= \frac{1}{2} (1+2\sqrt{2}i-2 + 1-2\sqrt{2}i-2)$ Show

$= \frac{1}{2} (-2) = -1$ ✓

(iii) In (ii), we have shown the result is true when $n=1$ and $n=2$.

Assume it is true for $n=k$ and $n=k+1$, and try to show it is true for $n=k+2$

i. Assume $T_k = \frac{1}{2} \left((1+i\sqrt{2})^k + (1-i\sqrt{2})^k \right)$

and $T_{k+1} = \frac{1}{2} \left((1+i\sqrt{2})^{k+1} + (1-i\sqrt{2})^{k+1} \right)$

RTP $T_{k+2} = \frac{1}{2} \left((1+i\sqrt{2})^{k+2} + (1-i\sqrt{2})^{k+2} \right)$

Now $T_{k+2} = 2T_{k+1} - 3T_k$

$= 2 \times \frac{1}{2} \left((1+i\sqrt{2})^{k+1} + (1-i\sqrt{2})^{k+1} \right)$

$- 3 \times \frac{1}{2} \left((1+i\sqrt{2})^k + (1-i\sqrt{2})^k \right)$

15 (b) (iii) (continued)

$$T_{k+2} = \frac{1}{2} \left[2(1+\sqrt{2}i)(1+\sqrt{2}i)^k - 3(1+\sqrt{2}i)^k + 2(1-\sqrt{2}i)(1-\sqrt{2}i)^k - 3(1-\sqrt{2}i)^k \right] \checkmark$$

$$= \frac{1}{2} \left[(1+\sqrt{2}i)^k (2+2\sqrt{2}i-3) + (1-\sqrt{2}i)^k (2-2\sqrt{2}i-3) \right]$$

$$= \frac{1}{2} \left[(1+\sqrt{2}i)^k (1+2\sqrt{2}i-2) + (1-\sqrt{2}i)^k (1-2\sqrt{2}i-2) \right] \checkmark$$

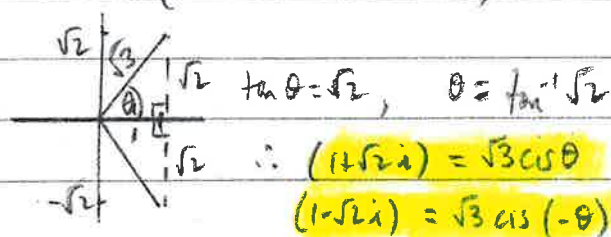
$$= \frac{1}{2} \left[(1+\sqrt{2}i)^k (1+\sqrt{2}i)^2 + (1-\sqrt{2}i)^k (1-\sqrt{2}i)^2 \right] \checkmark$$

$$= \frac{1}{2} \left[(1+\sqrt{2}i)^{k+2} + (1-\sqrt{2}i)^{k+2} \right] \checkmark$$

as required.

$\therefore$ If the result is true for $n=k, k+1$, it is also true for $n=k+2$, Since true for $n=1, 2$, by the process of Mathematical Induction, it is true for all $n \in \mathbb{Z}^+$

$$(iv) T_n = \frac{1}{2} \left[(1+\sqrt{2}i)^n + (1-\sqrt{2}i)^n \right]$$



$$\therefore T_n = \frac{1}{2} \left[(\sqrt{3} \cos \theta)^n + (\sqrt{3} \cos(-\theta))^n \right] \checkmark$$

$$= \frac{1}{2} (\sqrt{3})^n (\cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta))$$

$$= \frac{1}{2} (\sqrt{3})^n (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta)$$

$$= \frac{1}{2} (\sqrt{3})^n (2 \cos n\theta)$$

$$= (\sqrt{3})^n \cos n(\tan^{-1} \sqrt{2}) \checkmark$$

De Moivre

(16) (a) $I = \int_0^1 \sqrt{x-x^2} dx$

(1) $I = \int_0^1 \sqrt{\frac{1}{4} - (x-\frac{1}{2})^2} dx$

$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1}{4} - \frac{1}{4} \sin^2 \theta} \cdot \frac{1}{2} \cos \theta d\theta$

$= \frac{1}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$

$= \frac{1}{8} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$

$= \frac{1}{8} \left[\theta + \frac{1}{2} \sin 2\theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= \frac{1}{8} \left(\frac{\pi}{2} + 0 - (-\frac{\pi}{2} + 0) \right)$

$= \frac{\pi}{8}$

OR

$I = \int_0^1 \sqrt{x-x^2} dx$

Let $x = \sin^2 \theta$

$dx = 2 \sin \theta \cos \theta d\theta$

$I = \int_0^{\frac{\pi}{2}} \sqrt{\sin^2 \theta (1 - \sin^2 \theta)} \cdot 2 \sin \theta \cos \theta d\theta$

$= \int_0^{\frac{\pi}{2}} 2 (\sin \theta \cos \theta)^2 d\theta$

$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^2 2\theta d\theta$

$= \frac{1}{4} \int_0^{\frac{\pi}{2}} (1 - \cos 4\theta) d\theta$

$= \frac{1}{4} \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\frac{\pi}{2}}$

$= \frac{1}{4} \left[\frac{\pi}{2} - 0 - (0 - 0) \right]$

$= \frac{\pi}{8}$

Let $x - \frac{1}{2} = \frac{1}{2} \sin \theta$

$dx = \frac{1}{2} \cos \theta d\theta$

$\begin{cases} x=0, \theta = -\frac{\pi}{2} \\ x=1, \theta = \frac{\pi}{2} \end{cases}$

$\begin{cases} x=0, \theta = -\frac{\pi}{2} \\ x=1, \theta = \frac{\pi}{2} \end{cases}$

No marks for a substitution that led nowhere

(11) $I_n = \int_0^1 x^{n+\frac{1}{2}} \sqrt{1-x} dx$, integral $n \geq 0$.

Let $u = x^{n+\frac{1}{2}}$

$u' = (n+\frac{1}{2}) x^{n-\frac{1}{2}}$

$v' = (1-x)^{\frac{1}{2}}$

$v = -\frac{2}{3} (1-x)^{\frac{3}{2}}$

$I_n = \left[-\frac{2}{3} x^{n+\frac{1}{2}} (1-x)^{\frac{3}{2}} \right]_0^1 + \frac{2}{3} (n+\frac{1}{2}) \int_0^1 (1-x)^{\frac{3}{2}} x^{n-\frac{1}{2}} dx$

$= [0-0] + \frac{2n+1}{3} \int_0^1 \sqrt{1-x} (1-x) x^{n-\frac{1}{2}} dx$

$= \frac{2n+1}{3} \left(\int_0^1 \sqrt{1-x} (x^{n-\frac{1}{2}} - x^{n+\frac{1}{2}}) dx \right)$

$3 I_n = (2n+1) (I_{n-1} - I_n)$

$(2n+4) I_n = (2n+1) I_{n-1}$

$\therefore I_n = \frac{2n+1}{2n+4} I_{n-1}$

(11) $\int_0^1 x^n \sqrt{x-x^2} dx = \int_0^1 x^{n+\frac{1}{2}} \sqrt{1-x} dx = I_n$

$= \frac{(2n+1)}{(2n+4)} \times \frac{2(n-1)+1}{2(n-1)+4} \times I_{n-2}$

$= \frac{(2n+1)(2n-1)(2n-3) \cdots 5 \times 3}{(2n+4)(2n+2)(2n) \cdots 8 \times 6} \times \frac{8 I_0}{4 \times 2}$

$I_0 = \frac{\pi}{8}$ from (1)

$= \frac{(2n+1)(2n-1)(2n-3) \cdots 5 \times 3}{2^{n+2} (n+2)!} \times \frac{(2n)(2n-2) \cdots 4 \times 2}{2^n n!} \times \pi$

$= \frac{(2n+1)(2n)(2n-1)(2n-2) \cdots 5 \times 4 \times 3 \times 2 \times \pi}{2^{2n+2} (n+2)! n!}$

$= \frac{(2n+1)! \pi}{2^{2n+2} (n+2)! n!}$

(Continued)

OR

Using Mathematical Induction
(outline only here)

$$\text{RTP } I_n = \frac{(2n+1)! \pi}{2^{2n+2} (n+2)! n!}$$

$$\text{Base Case: } I_0 = \frac{1! \pi}{2^2 2! 0!} = \frac{\pi}{8}$$

$$\text{Assume } I_k = \frac{(2k+1)! \pi}{2^{2k+2} (k+2)! k!}$$

$$\text{Try to show } I_{k+1} = \frac{(2k+3)! \pi}{2^{2k+4} (k+3)! (k+1)!}$$

Relatively straightforward using the

$$\text{recurrence relation } I_{k+1} = \frac{(2(k+1)+1)}{(2(k+1)+4)} I_k$$

$$(b) \text{ Let } c = \cos \theta, s = \sin \theta$$

$$(i) \text{ By de Moivre, } (c+is)^{2k} = \cos 2k\theta + i \sin 2k\theta$$

Also

$$(c+is)^{2k} = c^{2k} + \binom{2k}{1} c^{2k-1} is - \binom{2k}{2} c^{2k-2} s^2 - \binom{2k}{3} c^{2k-3} is^3 + \dots$$

Equating Imaginary parts.

$$\sin 2k\theta = \binom{2k}{1} c^{2k-1} s - \binom{2k}{3} c^{2k-3} s^3 + \binom{2k}{5} c^{2k-5} s^5 - \dots$$

$$\therefore \frac{\sin 2k\theta}{\sin \theta \cos \theta} = \binom{2k}{1} c^{2k-2} - \binom{2k}{3} c^{2k-4} s^2 + \binom{2k}{5} c^{2k-6} s^4 - \dots$$

$$= \binom{2k}{1} (1-s^2)^{k-1} - \binom{2k}{3} (1-s^2)^{k-2} s^2 + \binom{2k}{5} (1-s^2)^{k-3} s^4 - \dots$$

Hence $\frac{\sin 2k\theta}{\sin \theta \cos \theta}$ is a polynomial in $\sin^2 \theta$.

$$(ii) \text{ For } k=4, \frac{\sin 8\theta}{\sin \theta \cos \theta} = \binom{8}{1} (1-s^2)^3 - \binom{8}{3} (1-s^2)^2 s^2 + \binom{8}{5} (1-s^2) s^4 - \binom{8}{7} s^6$$

$$= 8(1-3s^2+3s^4-s^6) - 56(s^2-2s^4+s^6) + 56(s^4-s^6) - 8s^6$$

$$= 8-24s^2+24s^4-8s^6-56s^2+112s^4-56s^6+56s^4-56s^6-8s^6$$

$$= -2(64s^6-96s^4+40s^2-4)$$

$$\text{For } 64s^6-96s^4+40s^2-4=0, \text{ let } z=2\sin \theta=2s$$

$$\text{So } 64s^6-96s^4+40s^2-4=0$$

$$\Rightarrow \sin 8\theta = 0, \text{ but } \sin \theta \neq 0, \cos \theta \neq 0$$

$$\therefore 8\theta = k\pi, \quad k \neq 4n, n \in \mathbb{Z}$$

$$\theta = \frac{k\pi}{8}$$

$$\text{Let } k = -3, -2, -1, 1, 2, 3$$

$$\theta = -\frac{3\pi}{8}, -\frac{\pi}{4}, -\frac{\pi}{8}, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}$$

$$\text{So } z = 2\sin(-\frac{3\pi}{8}), 2\sin(-\frac{\pi}{4}), 2\sin(-\frac{\pi}{8}), 2\sin(\frac{\pi}{8}), 2\sin(\frac{\pi}{4}), 2\sin(\frac{3\pi}{8})$$

$$\text{So } z = \pm 2\sin \frac{\pi}{8}, \pm \sqrt{2}, \pm 2\sin \frac{3\pi}{8}$$